

GOplan Manual

GOplan is an R package designed to manage animal breeding programs, including the breeding program of the core population and the whole breeding strategy of the crossbreeding system. Moreover, it allows optimization of the crossbreeding system.

The package has three main functions. Parameter “*prm_path*” is the full path of the input parameter file:

runCore(prm_path) is the function designed to assess various breeding schemes for the nucleus herds, you can compare the genetic progress as well as the monetary profit under different key elements in constructing populations such as the productive lifetime of breeding stocks, mating ratio, mating methods, the number of dams, cull rate, the method of breeding value estimation, the number of individuals selected for phenotyping or genotyping, and so on. It returns five aspects of results, the realized genetic progress, the final population mean of traits, the decline of genetic variance, the inbreeding increment, and the economic profit. Widely used BV estimation methods including BLUP, GBLUP, and ssGBLUP (single-step GBLUP) are supported by this function, which can be executed through calling the DMU software. Additionally, to improve the flexibility of mating methods, users can invoke external mating programs to generate their own mating plans by specifying the program's path.

runWhole(prm_path) designed to easily evaluate the performances of crossbred animals and the economic profit of complex crossbreeding programs with built-in frameworks. It allows the evaluation of different crossbreeding systems and can evaluate the effectiveness of crossbreeding program with different population structures by returning the predicted value of hybrids' phenotype and economic profit.

runOpt(prm_path) is used to search for an optimal combination of parameter values which maximizing the economic profits of crossbreeding programs. By defining some uncertain parameters and giving reasonable variation ranges, it uses the Bayesian Optimization method to draw up a superior solution under fixed iteration times.

Description of parameter file:

All needed information is determined by editing the parameter files. It is worth noting that the parameters you do not need should be specified by 0, and do not use any other separators except blank to separate different numbers, or it may cause unknown errors. Do not change the sequence of each parameter.

The parameter file contains six parts, each part is illustrated below.

Analyse Parameters

This part is used to determine some basic information when running all three functions.

parameters	value	explanation
QUICK	FALSE	#FALSE, TRUE
Method	blup	#blup, ssgblup, gblup
EarlySelect	FALSE	#FALSE, TRUE
Lit_pro	0	#the percentage of pre-selected litters, range from 0 to 1
Mate	rand	#Mate method: rand, MC or the path of extra program
maxF	999	#the maximum kinship coefficient between parents
Ctype	33	#type of crossbreeding: 1, 21, 22, 31, 32, 33, 41, 42
Nrep	50	#number of repetitions
Time	20	#time budget of breeding cycle
Ncores	10	#number of threads
nChr	18	#number of simulated chromosomes
nSnperChr	3000	#number of SNP per chr
nQtlPerChr	100	#number of QTL per chr
out_path		#path of output file

QUICK: when set to TRUE, the program would simplify the simulation of nucleus population by replacing the process of breeding value estimation with simulation. It would just calculates the first generation's estimated breeding value (EBV) (estimate the breeding value using DMU, then use the correlation between EBV and TBV as the accuracy of breeding value estimation), and use Cholesky method to simulate the other generations' breeding value according to the estimation accuracy. We assume TBV and EBV $\sim N(u, s^2)$, so the equation of simulating EBV is:

$$EBV_{sim} = u + r * (TBV - u) + \sqrt{1 - r^2} * e * s$$

Where EBV_{sim} is the vector of simulated EBV of all individuals, u is the mean value of all individuals' TBV, r is the accuracy (Pearson's correlation coefficient between TBV and EBV), TBV is the vector of true breeding value, e is a randomly generated vector that follows a normal distribution $\sim N(0,1)$, s is the standard deviation of TBV.

The QUICK option can save time and get a result quickly, if you just want to have a qualitative understanding of which breeding program is better and do not care about the exact profit, or when the **Method** is set to 'gblup' or 'ssgblup', we suggest set QUICK to TRUE.

Method: the program now supports three methods to evaluate the breeding value, and call DMU to run it. blup: best liner unbiased prediction, use only pedigree information; **ssgblup**: single-step blup, use both genomic and pedigree information; gblup: genomic blup, use only genomic information.

EarlySelect: FALSE, whether do pre-select for litters. If TRUE, GOplan would select some litters according to their litter index, which is the average of the father's breeding value and the mother's breeding value. Litter with high litter index would be selected.

Lit_pro: the the percentage of pre-selected litters, range from 0 to 1.

Mate: there are four options for users to choose, rand (random mating), HOMO: positive assortative mating, HETER: negative assortative mating, MC (minimal-coancestry mating, using annealing algorithm to find the solution); or you can specify the full path of your own mating program, while our package would write a file called “plans.txt” containing four columns, id of dam, id of sire, relationship coefficient and the expected breeding value of offspring out. See implementation information of MC in the additional file.

maxF: the maximum kinship coefficient between parents. Males and females with kinship coefficients greater than this value will not mate. When set to 999, males and females are mated without any restriction.

Ctype: crossbreeding type, see detail in Figure 1, 1 means single population, 2* means tow-way crossbreeding (B is the sire line, A is the dam line), 3* means three-way crossbreeding (C is the sire line, B is the sire of dam line, A is the dam of dam line), and 4* means four-way crossbreeding (D is the sire of sire line, C is the dam of sire line, B is the sire of dam line, A is the dam of dam line).

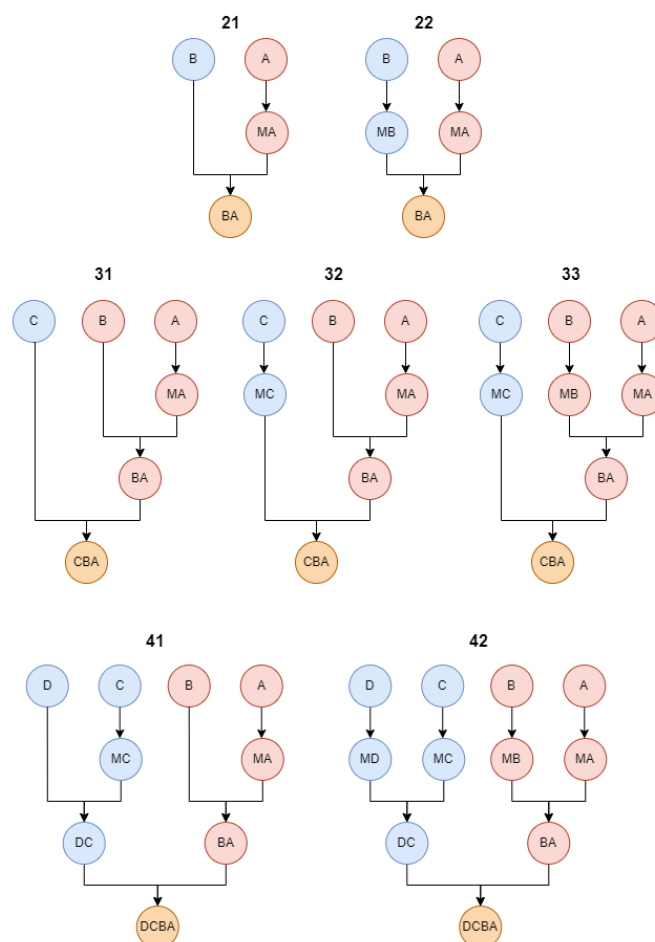


Figure1: the sketch map of different crossbreeding types.

Nrep: the number of repetitions. Final results would be the average of all repetitions.

Time: time budget of breeding cycle.

Ncores: number of threads. Our package supports multi threaded running. But please make sure that **Ncores** do not exceed the maximum available thread in your system.

nChr: number of simulated chromosomes. Change this parameter to adapt to different breeds.

nSnPPerChr: number of SNP per chromosome.

nQtlPerChr: number of QTL per chromosome.

out_path: path of output file, ended with “/”.

VARIABLES

This part can either be specified or ignored

The number of variables does not have limitation, but more variables means more combinations, which may cause long time running. We recommend to set less than three variables each time, one variable one row, with all the levels listed separated by blank.

Available variables for **runCore(prm_path)**: Ys, Yd, Sor, nfam, nfam_F, nfam_M, ngeno, nGeno_F, nGeno_M. Using nfam or ngeno means adjust nfam_F and nfam_M or nGeno_F and nGeno_M simultaneously.

Available variables for **runWhole(prm_path)**: Asec, Bsec, Csec, Dsec, YABd, YCDs, SorAB, SorCD, SorP

Optimize Parameters

This part used to modify the range of each optimize parameter. If you don't need some parameters, just delete the row of it. When use **runCore(prm_path)**, **runWhole(prm_path)**, ignore this section.

parameters	value	value	explanation
Pars	lower	upper	
Asec	2	6	#longevity of mutiplier A
Bsec	1	4	#longevity of mutiplier B
Csec	2	6	#longevity of mutiplier C
Dsec	1	4	#longevity of mutiplier D
YABd	2	6	#longevity of hybrid dam AB
YCDs	1	4	#longevity of hybrid dam CD
NAd	600	3000	#female size of nucleus A
NBd	300	1000	#female size of nucleus B
NCd	300	1000	#female size of nucleus C
NDd	300	1000	#female size of nucleus D

Population Structure

This part used to specify the structure of population in the breeding program.

parameters	value	value	value	value	explanation
------------	-------	-------	-------	-------	-------------

dam0	2				#age of dams when first born
sire0	2				#age of sires when first born
YABd	6				#longevity of hybrid dam AB
cull_AB	0	0	0		#cull rates of each age in AB
YCDs	4				#longevity of hybrid sire CD
cull_CD	0	0	0		#cull rates of each age in CD
SorAB	100				#Mating ratio when AB cross
SorCD	100				#Mating ratio when CD cross
SorP	100				#Mating ratio of terminal cross
ABSR	0.85				#survival rate of AB
CDSR	0.85				#survival rate of CD
PSR	0.85				#survival rate of products
N_P	0				#expected number of final products
nPoints	0				#number of selected points each iteration
nIter	0				#number of iterations
Breed	A	B	C	D	
n_female	500	500	500	0	#female size
Sor	50	50	50	0	#mate ratio
Ys	2	2	2	0	#longevity of sires
Yd	4	4	4	0	#longevity of dams
n_progeny	12	12	12	0	#litter size
SR	0.85	0.85	0.85	0	#survival rate of offspring
nSec	3000	0	0	0	#size of multiplier
Ysec	6	2	2	0	#longevity of multiplier
nPedG	3	0	0	0	#the number of generations traced for generating pedigree
nGenoG	3	0	0	0	#the number of generations traced for generating genotype file
nSel_F	2	0	0	0	#the number of female selected per litter
nSel_M	2	0	0	0	#the number of male selected per litter
nfam_F	2	2	2	0	#number of phenotyping females per litter
nfam_M	2	2	2	0	#number of phenotyping males per litter
nGeno_F	0	0	0	0	#number of genotyping females per litter
nGeno_M	0	0	0	0	#number of genotyping males per litter

Breeds Details

This part used to specify the information of each breed. You must list the information of all breeds.

parameters	value	value	value	value	value	explanation
Breedname	A					
cull_s	0					#cull rates of each age in sires
cull_d	0	0	0			#cull rates of each age in dams
cull_sec	0	0	0	0	0	#cull rates of each age in multipliers
trait	JZRL					#names of considering traits
SexLimit	0					#0: non-limiting trait, 1: trait that only for male, 2: trait only for female
mean	180					#means of traits
var	100					#phenotype variances of traits
heri	0.3					#heritability of traits
weigh	-1					#weight coefficients of traits
analyse	1					#analyse method
## phenotype covariance ##						
JZRL	1					
Breedname	B					
cull_s	0					
cull_d	0	0	0			
cull_sec	0	0	0	0	0	
trait	JZRL					
SexLimit	0					
mean	170					
var	100					
heri	0.3					
weigh	-1					
analyse	1					
## phenotype covariance ##						
JZRL	1					
Breedname	C					
cull_s	0					
cull_d	0	0	0			
cull_sec	0	0	0	0	0	

trait	JZRL					
SexLimit	0					
mean	160					
var	100					
heri	0.3					
weigh	-1					
analyse	1					
## phenotype covariance ##						
JZRL	1					

analyse: "1" means single trait, other same numbers mean multiple traits. For example, if there are three traits, T1, T2 and T3, T1 need to be estimated by single trait, while T2 and T3 need to be estimated using multi-trait model. Then we set analyse: 1, 2, 2.

phenotype covariance ##: this part determines the covariance of traits. For example, when there are three traits, T1, T2 and T3. Then it needs to be adjusted as below:

T1	1	0	0
T2	0	1	0
T3	0	0	1

Note: the column names are the same as the row names, though we omit them.

The parameter number of cull rate should always be one less than its corresponding longevity. For example, if the Yd of breed C is 4, then the cull_d of C must have three values (0, 0, 0). Because when parent exceed their longevity, they would be all culled.

Basic Economic Parameters

This part defines the economic information of the breeding program.

parameters	value	value	explanation
trait	0	JZRL	
jb_cost	906	1.5	#basic cost per individual
dam_cost	1500	0	#dam cost per individual
sire_cost	1000	0	#sire cost per individual
other_cost	10	0	#other cost each season
ind_sale	2040	0	#income of selling a product
cull_sale	1500	0	#income of culling
sire_sale	3000	0	# income of selling a sire
dam_sale	2000	0	# income of selling a dam

meas_cost	20		#phenotyping cost per individual
geno_cost	200		#genotyping cost per individual

trait: first column fixed with 0, means the initial value of all economic parameters. Then list the traits concerned, each trait corresponds to a column, means the monetary change when each trait changes one unit. The value can be negative.

jb_cost: the basic cost for raising an individual.

dam_cost: cost for keep a dam in the population each season.

sire_cost: cost for keep a sire in the population each season.

other_cost: total other cost each season.

ind_sale: income of selling a product.

cull_sale: income of culling a breeding stock.

sire_sale: income of selling a sire, a male who had been measured but not be selected can be sold as a sire.

dam_sale: income of selling a dam, a female who had been measured but not be selected can be sold as a dam.

meas_cost: phenotyping cost per individual.

geno_cost: genotyping cost per individual

Installation

```
install.packages("GOplan_0.1.0.tar.gz",repos = NULL)
```

Running Example & Output File

Example 1 :

```
runCore("example_prm/prm_Core.txt")
```

runCore() would return three files, coreOut.xlsx, coreOut_detail.xlsx and detailInfo.txt.

coreOut.xlsx has 9 columns listed below, and it is the results after the breeding process running after all breeding seasons.

mean_G / sd_G	Mean value and stand error of relative genetic progress
mean_Inb / sd_Inb	Mean value and stand error of inbreeding increase
Profit	The economic profit
meanPhenotype / sdPhenotype	Mean value and stand error of the predicted mean phenotype
Vg_meanPhenotype / Vg_sdPhenotype	Mean value and stand error of population genetic variance decrease of phenotype
var	The name of variation combination

coreOut_detail.xlsx records the information of each breeding season, and each type of information is saved in a separate sheet.

varg	Population genetic variance of each phenotype
acc	Estimation accuracy of each phenotype's breeding value
acc_index	Estimation accuracy of the aggregate selection index
index	The average aggregate selection index of population
dfInb	Inbreeding of population
pop0_G	Predicted mean phenotype

detailInfo.txt shows information about the population structure and breeding details under different variation combinations.

Here, we give a example of running breeding comparison with variables **Yd** in nucleus population. The parameter file was **prm_Core.txt** in folder `example_prm`: , and the results are shown below:

coreOut.xlsx :

mean_G	sd_G	mean_Inb	sd_Inb	Profit	meanJZRL	sdJZRL	Vg_meanJZRL	Vg_sdJZRL	var
0.12	0.01	0.03	0.01	5020030	158.40	1.11	-8.55	-1.76	2
0.11	0.01	0.02	0.00	5075439	160.42	1.64	-5.84	-1.54	4

coreOut_detail.xlsx :

mean-JZRL-2	sd-JZRL-2	mean-JZRL-4	sd-JZRL-4
0.705015559	0.03238569	0.683526646	0.047341928
<	>	>	varg
acc	acc_index	index	dfInb
pop0_G	+		

detailInfo.txt:

```

2
NAd: 500  NAs: 25  A_sor: 20
YAd: 2    YAs: 2
NAd0: 250      NAs0: 13
nfam_F: 2  nfam_M: 2
4
NAd: 500  NAs: 25  A_sor: 20
YAd: 4    YAs: 2
NAd0: 125      NAs0: 13
-

```

Example 2:

runWhole() would print out four files.

WholeOut.xlsx contains predicted phenotype of final hybrids and economic profit under different breeding programs.

*out.xlsx contains information about different subpopulations' phenotype in each breeding season.

detailInfo.txt is the same as described before.

p.csv is the P matrix used in gene flow method.

Here, we give a example of running crossbreeding comparison with variables **SorP** . The parameter file was **prm_Whole.txt** in folder `example_prm`:

runWhole("example_prm/prm_Whole.txt")

The results are shown below:

WholeOut.xlsx:

JZRL	Profit	SorP
160.7281	1118.256	50
159.8115	1121.354	100

***out.xlsx:**

[illegible]

Example 3:

```
runOpt(prm_path = "prm_Opt.txt")
```

The results of `runOpt()` are saved as `Opt_results.csv`.

	Asec	Bsec	Csec	YABd	NAd	NBd	NCd	profit	dob
1	6	2	1	2	694	447	300	671.0528	0
2	3	1	1	3	1816	742	535	1157.5452	0
3	4	4	3	2	795	451	417	641.3153	0
4	5	3	2	4	1708	318	362	1176.0509	0
5	3	4	1	6	1468	620	489	1192.6866	0
6	5	3	2	5	1946	559	813	1200.0366	0
7	4	2	4	3	1033	906	585	1153.2042	0
8	2	1	1	6	1673	886	718	1208.1767	0
9	6	4	4	2	882	699	775	1063.6187	0
10	2	3	2	5	1510	814	992	1199.4394	0
11	5	4	3	3	1157	374	667	1149.9696	0
12	2	1	4	5	1263	971	647	1195.4421	0
13	3	2	3	4	916	790	946	1178.6264	0
14	5	2	3	4	1397	533	873	1180.4610	0
15	6	4	4	6	1221	970	316	1202.0861	1
16	3	3	1	6	1354	727	695	1203.4154	1
17	6	1	4	6	1047	995	987	1207.3243	2
18	2	4	4	3	1388	775	791	884.1242	2
19	6	1	1	4	1877	884	429	1034.6625	3
20	5	4	4	6	1759	694	342	1204.9232	3
21	3	1	3	6	1191	513	592	1199.5786	4
22	3	1	4	5	1674	899	675	1199.4446	

Additional file

1 Implementation of minimal-coancestry mating (MC) using annealing algorithm

- a) Generate an initial mate plan stochastically, and calculate the overall coefficient of relationship: E_0 , set $T=1.0$, $N_{re} = 0$, $N_{eva} = 0$, where T is the annealing temperature, N_{re} is the number of plan's replace times, N_{eva} is the number of evaluation times;
- b) Change the plan by randomly selecting two mate pairs, and exchanging the dams, the new plan's E_i is equal with $E_0 + \delta$, where $\delta = a_{s1d2} + a_{s2d1} - a_{s1d1} - a_{s2d2}$, a_{s1d2} is the coefficient of relationship between sire 1 and dam 2 (Fig.2);
- c) If $\delta < 0$, accept the new plan, or accept it with the probability of $e^{(-\delta/T)}$;
- d) Update the T value with 10% reduction each time when either the plan has been replaced $10 \cdot n_{max}$ times or has already changed plan $100 \cdot n_{max}$ times, where n_{max} is the maximum between the number of dams and the number of sires;
- e) Stop if there is no improvement between the last and the second last change of T , or repeat the above steps.

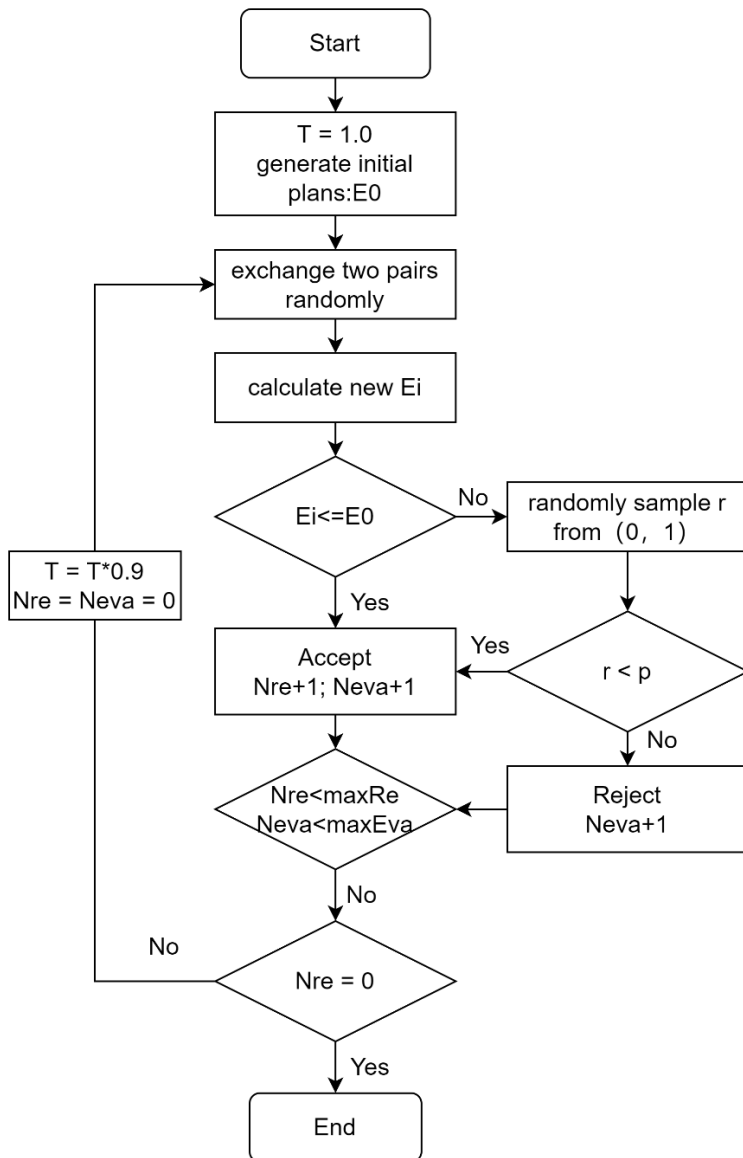


Fig. 1 A diagram illustrate the whole process of MC mate.

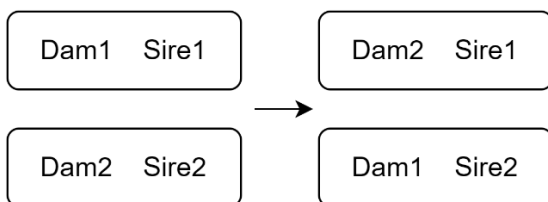


Fig. 2 Shows exchange of two mate pairs.

2 Calculation of monetary profit

In this package, we calculate the profit of each season separately and do not consider about the discount.

Main steps:

- a) Calculate the phenotype progress in time i , and figure out each economic part's value;
- b) Calculate the number of individuals of different types:

TYPE	CALCULATION METHOD
Weaning Number	The number of all dams * litter size
Culling Number	Total number of culled animals
Measuring Number	Total number of measured animals
Genotyping Number	Total number of genotyped animals
Sold Dams	Measuring Number of female - number of culled dams in core population or multiplier
Sold Sires	Measuring Number of male - number of culled sires in core population or multiplier
Product	The number of products' dams * litter size * survival rate
All Product	Weaning number * survival rate – max(Culling Number, (Measuring Number + number of culled animals of products' parents))

- c) Calculate the economic parts:

ECONOMIC PART	CALCULATION METHOD
All Basic Cost	Weaning Number * jb_cost'
All Sire Cost	Total number of sires * $sire_cost'$
All Dam Cost	Total number of dams * dam_cost'
All Measure Cost	Measuring Number * $meas_cost'$
All Genotyping Cost	Genotyping Number * $geno_cost'$
Income of selling products	All Product * ind_sale'
Income of Selling Dams	Sold Dams * dam_sale'
Income of Selling Sires	Sold Sires * $sire_sale'$
Income of Culling Animals	Culling Number * $cull_sale'$

- d) Profit = (Income of selling products + Income of Selling Sires + Income of Selling Dams + Income of Culling Animals) - (All Basic Cost + All Sire Cost + All Dam Cost + All Measure Cost + All Genotyping Cost + other_cost)